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**UPLOAD
PARTICIPANT**

**ANALYSE
PARTICIPANT**

**PARTICIPANTS
DATABASE**

Amblyopia, commonly referred to as “lazy eye,” is a neurodevelopmental visual disorder characterized by reduced visual acuity that cannot be explained by structural abnormalities of the eye alone. The condition typically develops during early childhood as a result of abnormal visual experience, including strabismus, refractive imbalance, or visual deprivation. Beyond reduced acuity, amblyopia has been associated with deficits in binocular vision, oculomotor control, visual attention, and higher-order visual processing.

Recent evidence suggests that amblyopia may also influence reading fluency and audiovisual integration processes. Efficient reading relies on accurate and temporally synchronized visual and auditory processing, requiring coordination between visual perception, eye movements, attention, executive functions, and language-related neural networks. Children with amblyopia often demonstrate slower reading speed and altered visual scanning patterns, potentially reflecting differences in the neural mechanisms supporting reading acquisition and fluency.

In particular, differences between left-eye and right-eye amblyopia may be relevant for understanding reading-related neural processing. Since reading depends on lateralized visual and language networks, asymmetric visual input may influence how information is integrated across hemispheres during reading tasks. Previous studies have suggested that amblyopia may affect functional connectivity, attentional engagement, and audiovisual synchronization, highlighting the importance of investigating neural activity alongside behavioral performance.

The current project focuses on the development of an integrated experimental and engineering platform designed to synchronize behavioral, physiological, and neurophysiological measurements during reading tasks. The system combines EEG, eye-tracking, and heart-rate acquisition with controlled visual and auditory experimental paradigms in order to investigate reading fluency, visual processing, and audiovisual integration in children with amblyopia.

The experimental session begins with the participant seated in front of a computer screen. The child is connected to a 64-channel EEG system, a Corsano wrist device for heart-rate and HRV recording, and a Tobii X3-120 eye-tracking system. After EEG and physiological sensors are connected, the eye-tracker calibration procedure is performed to ensure accurate gaze recording throughout the reading tasks.

1. Oral Reading - Erased Text

participants read the text aloud while letters gradually disappear from the screen over time. This condition combines active vocal reading with dynamic visual pacing.

2. Silent Reading - Erased Text

participants read the text silently while letters gradually disappear from the screen. No auditory pacing cues are presented during this task.

3. Silent Reading - Erased Text with Metronome

participants read silently while letters gradually disappear from the screen and rhythmic metronome clicks are simultaneously presented in the background. This condition is designed to examine the interaction between visual pacing and external auditory timing cues.

4. Silent Reading - Static Text with Metronome

participants read silently while the full text remains continuously visible on the screen. Rhythmic metronome clicks are presented throughout the task.

5. Silent Reading - Static Text

participants read silently while the full text remains continuously visible on the screen.

6. Resting state

In addition to the reading tasks, the experiment includes a two-minute resting-state baseline condition, during which the child looks at the screen without performing an active task. This baseline is used to characterize resting neural and physiological activity and to provide a comparison point for task-related EEG, eye-tracking, and heart-rate measures.

After each reading task, the child answers multiple-choice comprehension questions about the text. These questions are used to verify reading engagement and to provide behavioral measures of comprehension for each experimental condition.

UPLOAD PARTICIPANT

Code Implementation Notes:

All input fields are optional.

When a new subject folder is created, it is automatically named using the subject's ID number.

Inside the subject folder, separate subfolders are created for each uploaded task.

The subject's personal details (excluding task-related files) are stored in a centralized Excel table together with the records of all other subjects.

New Participant

Participant ID:

Participant Name:

Participant age:

Participant group: Control / Left Amblyopia/ Right Amblyopia

Notes:

EyeTracking.xlsx: *This field should support uploading multiple files.*
(settings: single excel file, export units, milliseconds, fixations)

acc.xlsx:

activity.xlsx :

heart_rate_variability.xlsx:

Files Upload

Task (Dropdown list): Oral Reading - Erased Text / Silent Reading - Erased Text / Silent Reading - Erased Text with Metronome / Silent Reading - Static Text with Metronome / Silent Reading - Static Text / Resting state

Task.ahdr:

Task.eeg:

Task.amrk:

Eprime.txt:

Reading comprehension assessment score (percentage):

ADD TASK

(open new task with the same upload options)

ADD PARTICIPANT

ANALYSE PARTICIPANT

All components displayed between "Task Level Analysis" and "Event Level Analysis" should appear only after the "Task Level Analysis" button has been clicked.

The tables displaying the following features should appear only after the button "GENERATE" is clicked after choosing task name.

The interface calculates these features only once — the first time the "GENERATE " button is pressed.

After the calculation is completed, the generated tables are saved in the subject's folder under the chosen task's folder. Each one under the number of the table.

On subsequent clicks of the " GENERATE" button, the table is loaded automatically from the saved file instead of being recalculated.

Clicking on any feature in the table will display a short explanation of the feature based on the project documentation.

The explanation should describe what the feature measures in practice, rather than its physiological interpretation.

Participants ID: (Dropdown list containing the ID numbers of all existing subjects)

Task Level Analysis

Task Name: (Dropdown list containing all existing tasks names associated with the selected subject)

GENERATE

Table 1: Eye Tracking Data

mean_fixation_duration
fixation_time_percentage
fixation_duration_variability
mean_saccade_duration
saccade_time_percentage
saccade_duration_variability
EyesNotFound_percentage
mean_EyesNotFound_duration
EyesNotFound_duration_variability
mean_pupil_diameter
pupil_diameter_variability
regression_percentage
mean_regression_distance
regression_duration_variability

Table 2: Physiological Data

mean_bpm
bpm_change_from_baseline
mean_respiration_rate
respiration_variability
respiration_change_from_baseline
mean_rmssd
rmssd_variability
rmssd_change_from_baseline
mean_si
si_variability
si_change_from_baseline

Table 3: EEG Data

mean_frontal_theta_power
theta_power_change_from_baseline
theta_power_variability
mean_occipital_alpha_power
alpha_power_change_from_baseline
alpha_power_variability
mean_theta_alpha_ratio
theta_alpha_ratio_change_from_baseline
theta_alpha_ratio_variability
mean_occipital_temporal_theta_plv
OT_plv_change_from_baseline
OT_plv_variability
mean_temporal_frontal_theta_plv
TF_plv_change_from_baseline
TF_plv_variability
mean_occipital_frontal_theta_plv
OF_plv_change_from_baseline
OF_plv_variability

Table 4: Quality control

left_valid_percentage
right_valid_percentage
combined_valid_percentage
mean_motion_magnitude
motion_variability
high_motion_percentage
motion_burst_count
Reading_comprehension_assessment_score

ANALYSE PARTICIPANT

Still within the "Analyze Participant" page, the user can select multiple task names, and multiple features at a time from a single list.

Once the feature is selected and the "Compare" button is clicked, a table is generated showing the selected feature values for each selected task.

The feature calculation can be performed at the time the "Compare" button is clicked, and the result does not need to be saved.

TASK COMPARISON

TASK NAMES: (Dropdown list of task names associated with the selected subject. The user should be able to select multiple task names.)

Eye Tracking Features: Dropdown list containing:

mean_fixation_duration
fixation_time_percentage
fixation_duration_variability
mean_saccade_duration
saccade_time_percentage
saccade_duration_variability
EyesNotFound_percentage
mean_EyesNotFound_duration
EyesNotFound_duration_variability
mean_pupil_diameter
pupil_diameter_variability
regression_percentage
mean_regression_distance
regression_duration_variability

Physiological Features: Dropdown list containing:

mean_bpm
bpm_change_from_baseline
mean_respiration_rate
respiration_variability
respiration_change_from_baseline
mean_rmssd
rmssd_variability
rmssd_change_from_baseline
mean_si
si_variability
si_change_from_baseline

EEG Features: Dropdown list containing:

mean_frontal_theta_power
theta_power_change_from_baseline
theta_power_variability
mean_occipital_alpha_power
alpha_power_change_from_baseline
alpha_power_variability
mean_theta_alpha_ratio
theta_alpha_ratio_change_from_baseline
theta_alpha_ratio_variability
mean_occipital_temporal_theta_plv
OT_plv_change_from_baseline
OT_plv_variability
mean_temporal_frontal_theta_plv
TF_plv_change_from_baseline
TF_plv_variability
mean_occipital_frontal_theta_plv
OF_plv_change_from_baseline
OF_plv_variability

Quality control Features: Dropdown list containing:

left_valid_percentage
right_valid_percentage
combined_valid_percentage
mean_motion_magnitude
motion_variability
high_motion_percentage
motion_burst_count

COMPARE

ANALYSE PARTICIPANT

All components displayed after "Event Level Analysis" should appear only after the corresponding button has been clicked.

The table generated after pressing the "Generate" button should be calculated only once during the first execution and then saved under the selected subject's folder, under the chosen task's folder, using the identifier corresponding to the event type.

On subsequent clicks, the table should be loaded directly from the database instead of being recalculated.

Clicking on any feature in the table will display a short explanation of the feature based on the project documentation.

The explanation should describe what the feature measures in practice, rather than its physiological interpretation.

EVENT LEVEL ANALYSIS

TASK NAMES: (Dropdown list of task names associated with the selected subject)

EVENT TYPE: dropdown list containing:

Long Fixation Events
Regression Events
EyesNotFound Bursts

GENERATE

Table 5- for Long Fixation Events / 6- Regression Events / 7- EyesNotFound Bursts

number_of_events
mean_event_duration
event_duration_variability

Table 8- for Long Fixation Events / 9- Regression Events / 10- EyesNotFound Bursts

	mean	standart_deviation	variance
EEG_feature			

EEG_feature:

frontal_theta_power
occipital_alpha_power
frontal_theta_occipital_alpha_ratio
theta_PLV_occipital_temporal
theta_PLV_temporal_frontal
theta_PLV_occipital_frontal

PARTICIPANTS DATABASE

*The Participants
Database page should
support:
- searching for a
specific participant
using participant ID
- filtering participants
by participant group*

*If no search or filter is
applied, all existing
participants should be
displayed in the table by
default.*

*Clicking on a
participant ID within
the table should
automatically open the
"Analyse Participant"
page for the selected
participant.*

PARTICIPANTS DATABASE

Search by ID:

Filter by group: *dropdown list containing:*

Control

Left amblyopia

Right amblyopia

Participant ID	Name	Group	Age	Existing task